CYBERSECURITY INTERNSHIP | WEEK 1 PROJECT REPORT

WK1-PM1

Cybersecurity Lab Setup using VirtualBox and Kali Linux

Project Report

Task Code	WK1-PM1
Domain	Cybersecurity
Tools	Oracle VirtualBox and Kali Linux
Programme	Week 1 Cybersecurity Internship Task

For educational use in an isolated virtual lab environment.

Contents

1. Introduction 1

2. Objective 1

3. Tools and Technologies Used 2

4. Lab Setup and Configuration 2

5. Practical Implementation 4

6. Screenshots and Observations 5

7. Testing and Results 6

8. Challenges and Troubleshooting 7

9. Learning Outcomes 7

10. Conclusion 8

1. Introduction

A cybersecurity testing lab is an isolated environment in which security tools and techniques can be learned and practised without affecting real systems or networks. Most professional security work, including penetration testing, network analysis and vulnerability assessment, begins in a controlled environment of this kind.

This report documents the first practical task of the internship, **WK1-PM1**. The task is to prepare the foundation of such a lab by running Kali Linux as a virtual machine inside Oracle VirtualBox on a host computer, and to confirm that the virtual machine can reach the internet.

A virtual lab is preferred over a physical one because it is inexpensive, easy to reset and safe to experiment in. The virtual machine is separate from the host operating system, so mistakes made inside the lab do not damage the host. Several machines can also be run side by side and connected through virtual networks.

> **Note on sources.** This report is based on the task requirements as described by the intern and on three screenshots taken from the lab. The original project instruction PDF was not available when this report was prepared. Statements are marked as **Verified** only where a screenshot supports them. Everything else is listed as a task requirement.

2. Objective

The objective of this task is to set up a working Kali Linux virtual machine in Oracle VirtualBox that can serve as a base for cybersecurity and ethical hacking practice. Kali Linux is a Debian-based distribution that is widely used for security testing and includes many security tools.

The specific goals of the task are:

- Install and use Oracle VirtualBox as the virtualization platform on the host computer.
- Run Kali Linux as a virtual machine in VirtualBox.
- Connect the virtual machine to a NAT Network with the subnet **10.0.0.0/24**.
- Configure Kali Linux with the IP address **10.0.0.2/24**.
- Confirm internet connectivity from Kali Linux using a network test such as `ping`.

3. Tools and Technologies Used

Tool / Technology	Purpose in this lab	Evidence in screenshots
Oracle VirtualBox	Hypervisor that hosts and runs the Kali Linux virtual machine.	VirtualBox Manager window (Screenshot 1).
Kali Linux	Security-focused Linux distribution used as the lab machine. The VM description reads Kali Rolling (2026.3) x64.	VM details (Screenshot 1); Kali desktop (Screenshot 2).
Host operating system	The computer on which VirtualBox runs. The host appears to be Windows: the VM audio host driver is listed as Windows DirectSound and the window frame is a Windows style. The exact Windows version is not shown.	Screenshot 1 and 2.
Terminal (Kali)	Command-line interface used to run network commands.	Terminal window (Screenshot 3).
ping command	Sends ICMP echo requests to test reachability and measure response time.	Output from `ping google.com` (Screenshot 3).

4. Lab Setup and Configuration

4.1 VirtualBox environment

Oracle VirtualBox is installed on the host computer and its Manager window lists the virtual machines available. Two machines appear in the list: **Redhat**, which is powered off, and **kali-linux-2026.3-virtualbox-amd64**, which is running. Only the Kali Linux machine is used in this task.

4.2 Kali Linux virtual machine

The details panel of the Kali Linux virtual machine in VirtualBox Manager shows the following settings.

Setting	Value shown in VirtualBox Manager
Name	kali-linux-2026.3-virtualbox-amd64
Operating system type	Debian (64-bit)
Base memory	2048 MB
Processors	2
Boot order	Hard Disk, Optical
Acceleration	Nested Paging, PAE/NX, KVM Paravirtualization

Setting	Value shown in VirtualBox Manager
Video	128 MB video memory, VMSVGA graphics controller
Storage	SATA virtual disk kali-linux-2026.3-virtualbox-amd64.vdi (Normal, 80.09 GB); optical drive on IDE is empty
Audio	Host driver Windows DirectSound, controller ICH AC97
USB	OHCI controller, 0 device filters
Shared folders	None

4.3 NAT Network configuration

Adapter 1 of the virtual machine is an **Intel PRO/1000 MT Desktop** adapter attached to a **NAT Network** named **NatNetwork1**. A NAT Network lets virtual machines share a private subnet and reach external networks through the host, while the machines themselves are not directly reachable from outside. This suits a lab, because the lab machines are isolated but still have internet access for updates and downloads.

The task requires this NAT Network to use the subnet **10.0.0.0/24** and Kali Linux to be given the address **10.0.0.2/24**. The VirtualBox screenshot confirms that the adapter is attached to a NAT Network, but it does not show the subnet range of NatNetwork1.

4.4 Task requirements and verification status

The table below separates what the task asks for from what the supplied screenshots actually confirm.

Requirement / item	Status	Basis
Kali Linux VM running in VirtualBox	**Verified**	VM state shown as Running (Screenshot 1); Kali desktop visible (Screenshot 2).
VM attached to a NAT Network	**Verified**	Adapter 1 shows NAT Network 'NatNetwork1' (Screenshot 1).
Internet connectivity from Kali	**Verified**	Successful ping replies from google.com (Screenshot 3).
NAT Network subnet 10.0.0.0/24	Required, not shown	Subnet range is not visible in any screenshot.
Kali IP address 10.0.0.2/24	Required, not shown	No screenshot of `ip a` or similar output was provided.
Shared folders	Not configured	VirtualBox shows 'None'.
Clipboard, drag-and-drop, snapshots	Not evidenced	No screenshot covers these, so no claim is made.

5. Practical Implementation

This section describes the steps of the lab as far as they are supported by the screenshots.

5.1 Launching Kali Linux in VirtualBox

- **Open VirtualBox Manager.** The Kali Linux virtual machine is selected in the machine list.
- **Start the virtual machine.** Its status changes to Running, and the preview thumbnail in the details panel shows the Kali desktop.
- **Reach the desktop.** The Kali Linux desktop loads inside the VirtualBox window, titled 'kali-linux-2026.3-virtualbox-amd64 [Running]'. It shows Home, File System and Trash icons, a top panel with four workspaces, and the Kali wallpaper.

5.2 Network connectivity testing with ping

The `ping` command sends small ICMP echo request packets to a target and reports each reply together with its round-trip time. A steady stream of replies shows that the machine can resolve the name, send traffic out and receive answers.

- **Open a terminal.** A terminal window titled 'kali@kali: ~' is opened on the Kali desktop.
- **Run the test.** The command `ping google.com` is entered at the prompt.
- **Observe the replies.** Kali resolves the name to an IP address and receives replies, with the sequence number increasing by one each time. The results are analysed in Section 7.

Setting the NAT Network subnet to 10.0.0.0/24 and assigning 10.0.0.2/24 to Kali Linux are part of the task requirements. Their configuration steps are not covered by the screenshots, so they are not described here as completed activities.

6. Screenshots and Observations

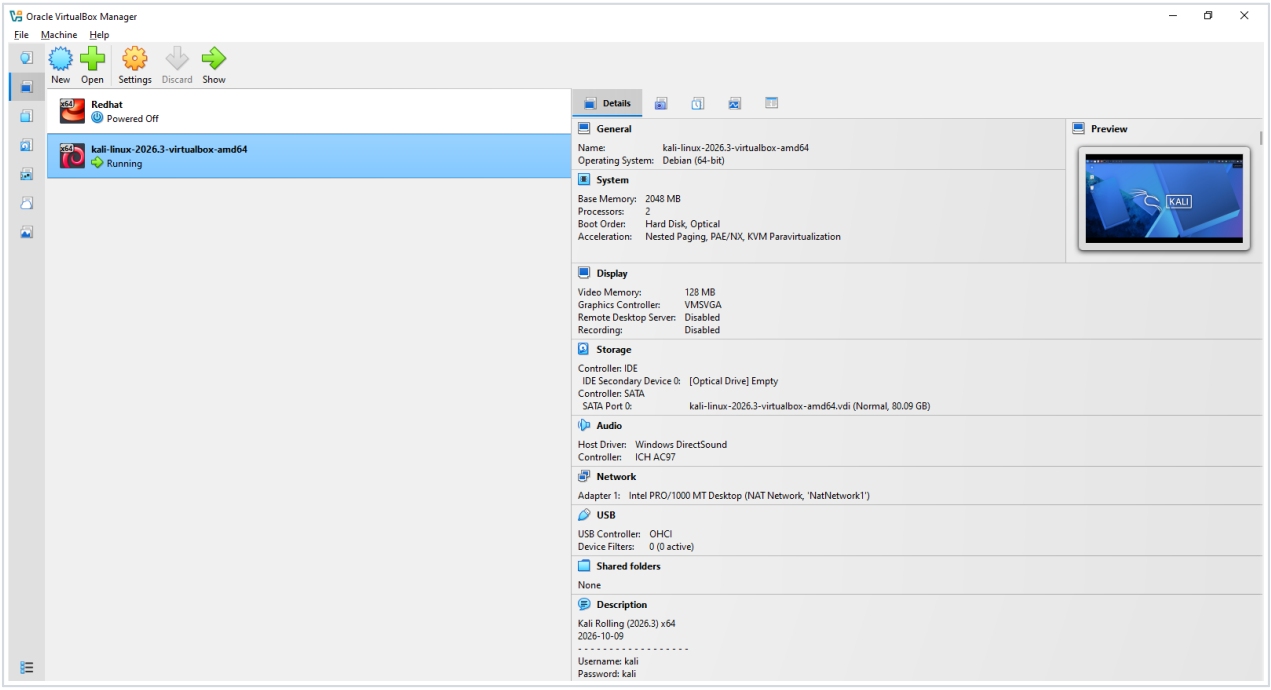


Figure 1: VirtualBox Manager showing the Kali Linux virtual machine running

The machine *kali-linux-2026.3-virtualbox-amd64* is highlighted with the status Running. The details panel shows 2048 MB memory, 2 processors, an 80.09 GB virtual disk, and network Adapter 1 connected to the NAT Network 'NatNetwork1'. The Shared folders entry reads None. The second machine, Redhat, is powered off.

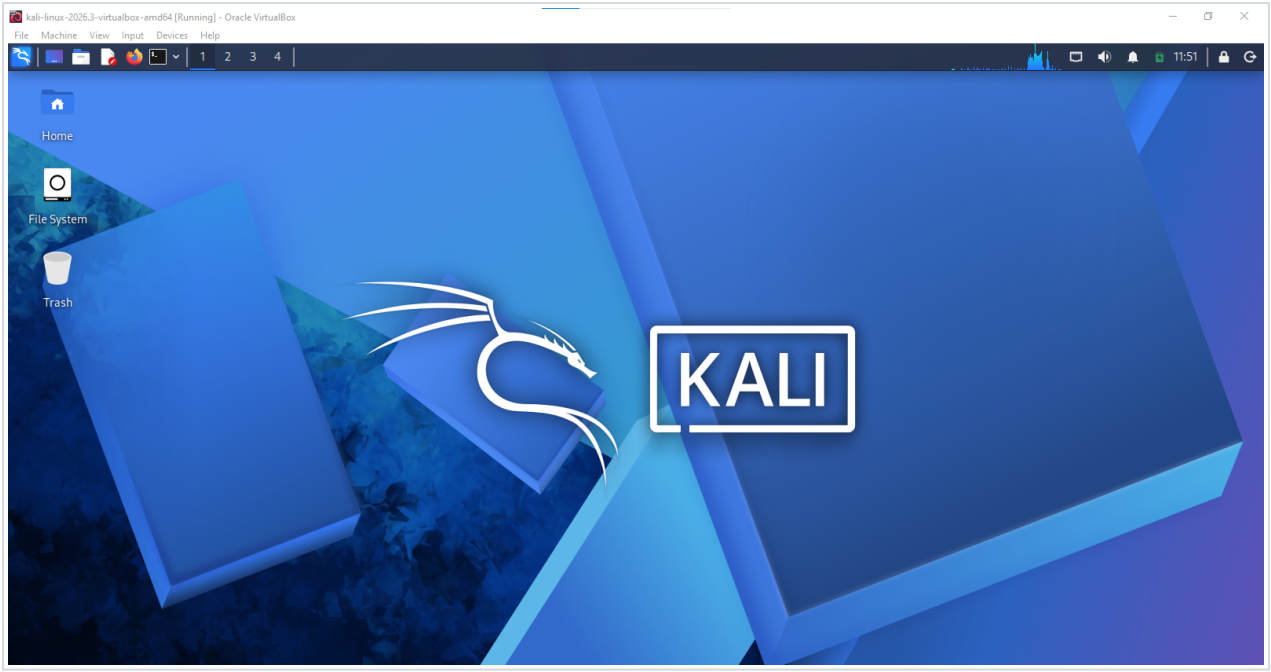


Figure 2: Kali Linux desktop environment

The Kali Linux desktop has loaded inside the VirtualBox window. The panel at the top holds the application menu, file manager, web browser, terminal launcher and four numbered workspaces. The clock shows 11:51. This confirms that the guest operating system booted successfully and is usable.

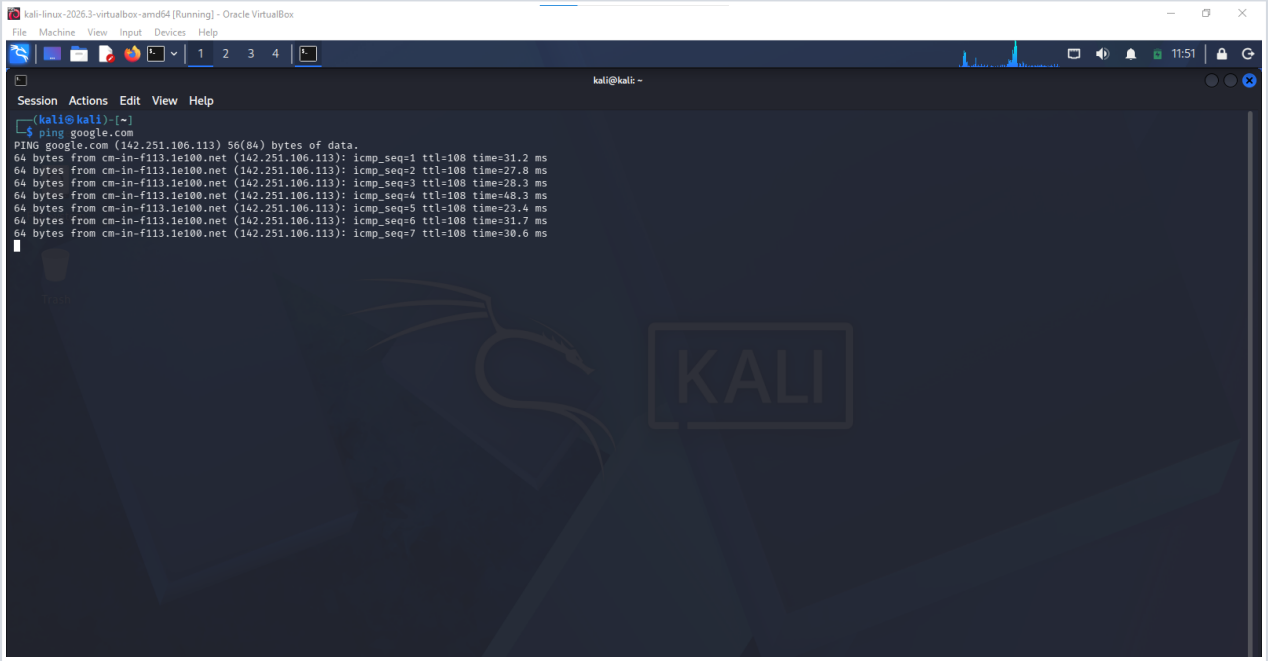


Figure 3: Terminal showing successful ping responses from google.com

The terminal prompt *kali@kali* shows `ping google.com` with replies from 142.251.106.113 for sequence numbers 1 to 7. Each reply is 64 bytes with a TTL of 108. The test was still running when the screenshot was taken, so the final ping statistics are not displayed.

7. Testing and Results

The connectivity test was carried out from the Kali Linux terminal against **google.com**. The command returned replies immediately and without visible gaps, which shows that the virtual machine has working internet access through its NAT Network adapter.

Measurement	Observed value (Screenshot 3)
Command	`ping google.com`
Name resolved to	142.251.106.113 (cm-in-f113.1e100.net)
Packet size	56 bytes of data, 84 bytes with headers; replies of 64 bytes
Replies seen	7 (icmp_seq 1 to 7, in unbroken sequence)
TTL	108 in every reply
Round-trip time	Minimum 23.4 ms, maximum 48.3 ms, average about 31.6 ms (calculated from the 7 visible replies)

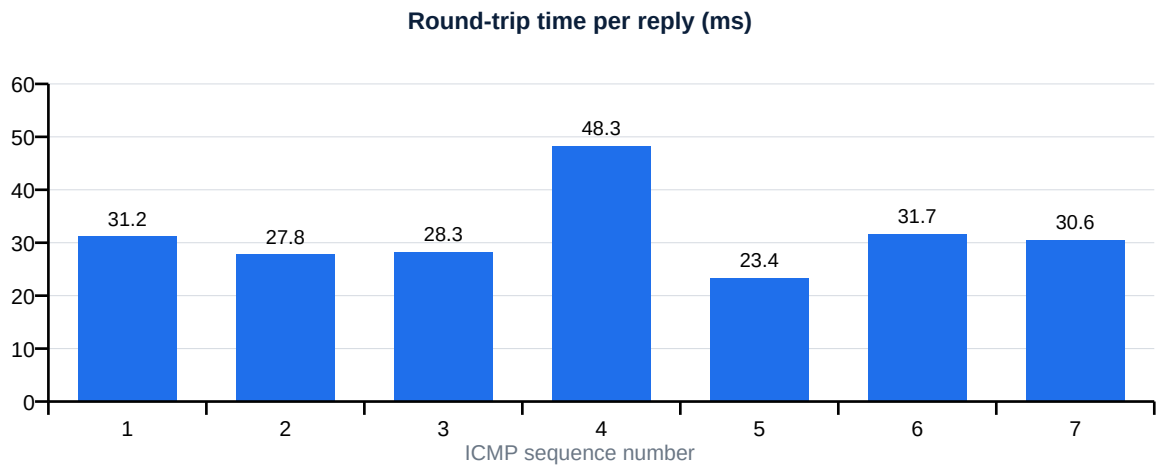


Figure 4: Round-trip time of the seven ping replies visible in Figure 3

Interpretation

- **Name resolution works.** The name google.com was converted to an IP address, so the virtual machine can reach a DNS service.
- **Outbound traffic works.** Echo replies came back from the external host, so packets leave the lab through the NAT Network and return.
- **Response times are consistent.** Most replies took between 23 and 32 ms, with one higher value of 48.3 ms. This is normal variation on an internet path.
- **No loss is visible.** All sequence numbers from 1 to 7 are present. A packet loss percentage cannot be quoted because the ping summary was not captured.

8. Challenges and Troubleshooting

No setup or network problems were reported for this task, and none are visible in the screenshots. For that reason, no challenges or fixes are listed here. Reporting only what actually happened keeps this record accurate.

9. Learning Outcomes

- **Virtualization.** Understanding how a hypervisor such as VirtualBox runs a complete guest operating system on a host, with its own allocated memory, processors and virtual disk.
- **Kali Linux.** Basic experience with starting Kali Linux, using its desktop and opening a terminal.
- **Virtual networking.** Understanding how a virtual network adapter attached to a NAT Network lets a guest reach the internet while remaining on a private subnet.
- **Connectivity testing.** Using `ping` to test connectivity and reading its output: resolved address, sequence numbers, TTL and round-trip time.
- **Documentation.** Recording a lab clearly, and separating verified results from requirements.

10. Conclusion

A Kali Linux virtual machine is running in Oracle VirtualBox, and its desktop loads correctly. Its network adapter is attached to a NAT Network, and a ping test to google.com returned successful replies with response times of roughly 23 to 48 ms. This confirms that the lab has working internet access, which is the main outcome the task asks for.

The required subnet 10.0.0.0/24 and the Kali address 10.0.0.2/24 are not visible in the supplied screenshots. A screenshot of the NAT Network settings in VirtualBox and of the output of `ip a` in Kali Linux would complete the evidence for those two items.

The lab can now be used as the base environment for the practical cybersecurity tasks that follow in the internship.